

1. Lecture Objectives

- By the end of this lecture, students will be able to:
 - Clearly define what a **software design pattern** is
 - Explain **why design patterns exist**
 - Distinguish between **design patterns, algorithms, and frameworks**
 - Understand the **origin and philosophy** of patterns
 - Identify **design problems** that patterns aim to solve
 - Recognize **benefits and dangers** of using patterns